

Powers and Exponents

Lesson 6-1

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

In 2^3 , the small 3 means multiply 2 by itself 3 times: $2 \times 2 \times 2 = 8$

Exponent

A small number that tells how many times to multiply the number by itself.

In 5^2 , the base is 5 — it is the number being multiplied: $5 \times 5 = 25$

Base

The number that gets multiplied by itself.

$10^3 = 10 \times 10 \times 10 = 1,000$ — read as '10 to the third power' or '10 cubed'

Power

A number written with a base and an exponent, like 2^3 .

Evaluate 3^4 : write $3 \times 3 \times 3 \times 3$, then multiply step by step: $9 \times 9 = 81$

Evaluate

To find the value of an expression.

In $3x$, the 3 is the coefficient — if $x = 4$, then $3x = 3 \times 4 = 12$

Coefficient

The number in front of a letter, like the 3 in $3x$.

$4x$ and $2x$ are like terms (both have x); $4x$ and $4x^2$ are NOT like terms (different powers)

Like terms

Terms with the same letter, like $2x$ and $5x$.

Key Ideas & Notes

- You're a sound engineer at a music studio.
- Each time you flip a switch, the volume doubles.
- The starting volume is 1 unit.
- After flipping the switch 3 times, the volume is $2^3 = 2 \times 2 \times 2 = 8$ units — that's 8 times louder!
- Evaluate each exponential expression. Write the repeated multiplication and find the value.

Think About It

- What happens to the volume each time you flip a switch?
- Why is 2^3 equal to 8, not 6?
- How is repeated multiplication different from repeated addition?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

In the sound studio, the volume doubles each time you flip the switch. Why is the volume after 3 flips $2^3 = 8$ and not $2 \times 3 = 6$?

Level 1 support

Start here: The exponent counts how many times we multiply, not what we add.

After 3 flips the volume is ____ because we multiply 2 three times.

Después de 3 cambios el volumen es ____ porque multiplicamos 2 tres veces.

It is not 6 because doubling means we ____, not add.

No es 6 porque duplicar significa que ____, no sumar.

Word bank: Exponent Base Power Evaluate

Listen for: Students connect each switch flip to one more factor of 2 and explain that $2^3 = 2 \times 2 \times 2 = 8$, distinguishing repeated multiplication from repeated addition (which would give 6).

Level 2

If the volume tripled instead of doubled each flip, how would you write the volume after 3 flips, and how many times louder would that be than the doubling case?

- Tripling gives ____ = ____, which is ____ times the doubling result because ____.
- Changing the base from 2 to 3 matters more than changing the exponent because ____.

EXPLORE

Look at the table: when the exponent goes up by 1 (like from 2^3 to 2^4), what happens to the value, and why?

Level 1 support

Start here: Each time the exponent grows by 1, the value is multiplied by the base one more time.

When the exponent increases by 1, the value is multiplied by ____.

Quando el exponente aumenta en 1, el valor se multiplica por ____.

This happens because the base is used as a factor ____ more time.

Esto pasa porque la base se usa como factor ____ vez más.

Word bank: Exponent Base Power Evaluate

Listen for: A strong answer states that the value gets multiplied by the base (e.g., $2^3 = 8$ becomes $2^4 = 16$, times 2) and ties it to adding one more factor, using the words base and exponent.

Level 2

Compare 2^5 and 5^2 . Both use the digits 2 and 5, so why are they not equal? Generalize: does the bigger exponent or the bigger base have more effect on the value?

- $2^5 = \underline{\hspace{1cm}}$ but $5^2 = \underline{\hspace{1cm}}$, so they are not equal because ____.
- In general, a larger ____ has more effect because ____.

CONNECT

Each recording layer doubles the file size, modeled by 4×2^n . Why does the file size grow so fast as you add layers?

Level 1 support

Start here: Doubling repeatedly is exponential growth, so the size explodes after only a few layers.

After 5 layers the file is ____ MB because $4 \times 2^5 =$ ____.

Después de 5 capas el archivo es ____ MB porque $4 \times 2^5 =$ ____.

It grows quickly because each layer ____ the size.

Crece rápido porque cada capa ____ el tamaño.

Word bank: **Power** **Base** **Exponent** **Evaluate**

Listen for: Students evaluate $4 \times 2^5 = 4 \times 32 = 128$ MB and explain that exponential (doubling) growth multiplies the size each step, unlike adding a fixed amount.

Level 2

Critique this claim: 'After 10 layers the file is just twice as big as after 5 layers.' Is that true? Justify with the powers involved.

- The claim is ____ because 2^{10} compared to 2^5 is ____, not just doubled.
- Exponential growth means going from 5 to 10 layers multiplies the size by ____.

Guided Examples

Example 1

What is the value of 4^3 ?

Solution: $4^3 = 4 \times 4 \times 4 = 64$.

Answer: A. 64

Example 2

What is the value of 6^2 ?

Solution: $6^2 = 6 \times 6 = 36$.

Answer: A. 36

Example 3

Which expression shows 5^3 as repeated multiplication?

Solution: 5^3 means use 5 as a factor 3 times: $5 \times 5 \times 5$. The exponent tells how many times to multiply, not what to multiply by.

Answer: A. $5 \times 5 \times 5$

Write About the Math

The Writing Revolution

I can explain my work using the words exponent, base, power, and evaluate.

1. Kernel Sentence subject + verb

Model: Exponent is a small number that tells how many times to multiply the number by itself.

Exponente es un número pequeño que dice cuántas veces multiplicar el número por sí mismo.

Write a kernel sentence about exponent. Use a subject and a verb.

Escribe una oración base sobre exponente. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Exponent matters in math

Exponente importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Exponent matters in math because ____.

Exponente importa en matemáticas porque ____.

but
pero

Exponent matters in math, but ____.

Exponente importa en matemáticas, pero ____.

so
entonces

Exponent matters in math, so ____.

Exponente importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about exponent.
Di un hecho verdadero sobre exponent.

Exponent ____.

Question *Pregunta*

Ask a question about exponent.
Haz una pregunta sobre exponent.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about exponent.
Muestra entusiasmo sobre exponent.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with exponent.
Dile a un compañero qué hacer con exponent.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

After 3 flips the volume is ____ because we multiply 2 three times.

Después de 3 cambios el volumen es ____ porque multiplicamos 2 tres veces.

It is not 6 because doubling means we ____, not add.

No es 6 porque duplicar significa que ____, no sumar.

When the exponent increases by 1, the value is multiplied by ____.

Cuando el exponente aumenta en 1, el valor se multiplica por ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which expression shows 5^3 as repeated multiplication?

A. $5 \times 5 \times 5$

B. 5×3

C. $5 + 5 + 5$

D. $3 \times 3 \times 3 \times 3 \times 3$

Show your work:

2. Evaluate: $3 \times (4 + 2)^2$

A. 108

B. 54

C. 36

D. 144

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A bacteria colony doubles every hour. Starting with 1 bacterium, write a power expression for the number after 6 hours. Explain why exponential growth is so much faster than adding the same number each hour.

Sentence starter: After 6 hours there are ____ bacteria because $2^6 =$ _____. If it grew by adding 2 each hour instead, there would only be ____ bacteria. Exponential growth is faster because _____.

Show your work:

Reflect — Exit Ticket

What is the value of 3^4 ?

- A. 81
- B. 12
- C. 34
- D. 64

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. $5 \times 5 \times 5 - 5^3$ means use 5 as a factor 3 times: $5 \times 5 \times 5$. The exponent tells how many times to multiply, not what to multiply by.
2. **Try It 2:** A. $108 - 4 + 2 = 6$. $6^2 = 36$. $3 \times 36 = 108$.
3. **Exit Ticket:** A. $81 - 3^4 = 3 \times 3 \times 3 \times 3 = 81$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Exponent is a small number that tells how many times to multiply the number by itself.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).